Raphael Huang – Curriculum Vitae

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Research Interests

FPGA/ASIC design and verification; formal verification of hardware systems; hardware security, side-channel analysis; VLSI, place-and-route algorithms; heterogeneous computing for ML.

Education

University of California, Santa Cruz — September 2025 - Present

Master of Science in Computer Science and Engineering
Expected Graduation: December 2026

University of California, Santa Cruz — September 2021 - June 2025

Bachelor of Science, *magna cum laude* in Computer Engineering
GPA: 3.93

Professional Experience

Graduate Teaching Assistant, UC Santa Cruz, Baskin Engineering — September 2025 - Present

- Coordinated logistical and technical aspects of a ~150 student Introduction to Logic Design course.
- Oversaw the creation of homework and exams, ensuring consistent evaluation of student learning.
- Lectured and held lab sections, debugging large structural Verilog designs in Vivado one-on-one with students.

Connected and Automated Vehicle R&D Intern, Argonne National Laboratory — June 2025 - August 2025

- Transitioned LARS (Longitudinal Automated Replay System), a real-time vehicle energy analytics tool, from Simulink to a modular ROS 2 stack, enabling rapid cross-vehicle deployment.
- Validated the system through simulation and on-road testing on a Cadillac LYRIQ, implementing an Iterative Learning Control algorithm that reduced root-mean-square tracking error by ~65%.
- Delivered an accessible platform to support research and workforce development for the collegiate EcoCAR EV Challenge and the broader CAV community [R2].

Instructional Design Research Assistant, UC Santa Cruz, Hardware Systems Collective — Jan - July 2025

- Redesigned the curriculum for CSE 100, Introduction to Logic Design, focusing on accessible, equitable, and pedagogically-sound coursework.
- Integrated open-source EDA tools such as Yosys and Verilator to enabled remote completion of coursework, enabling enrollment to scale to over 140 students per quarter (previously ~80).

Lead Instructor, iDTech Camps, Stanford University — July - August 2024

- Led 11 instructors teaching AI/ML in Python and VEX Robotics to 100+ students while adapting curriculum for a pilot program for international learners from China and Latin America.

Research and Projects

Characterizing FPGA Power Side-Channel Recovery Sensors — January 2026 - Present

- Building a framework to quantify the granularity and dynamic range of different delay-line implementations for FPGA **power side-channel recovery** attacks.
- Running trace collection on **Zynq Ultrascale** architecture and applying statistical analysis techniques (e.g. t-tests, normality tests) to create taxonomy of various sensor designs.

Runtime Reconfigurable LUT-based Logic Gate Networks on FPGAs January 2026 - May 2026

- Developed a nanosecond-scale **neural network** inference architecture by mapping differentiable lookup-table networks to reconfigurable FPGA overlays [R4].
- Utilized **PYNQ** architecture to bypass hardware synthesis and bitstream-loading bottlenecks.

Formal Methods for VLSI Timing Optimization September 2025 - January 2026

- Developing a formal methods approach to correct **Static Timing Analysis (STA)** violations by integrating a Z3 OMT solver with the **OpenROAD** open-source ASIC design flow.
- Created **Python** and **Tcl** scripts to parse timing reports, generate logical constraints for optimal cell resizing, and programmatically modify netlists [R1].

Vivaldi: A SystemVerilog FPGA Audio Synthesizer January - March 2025

- Wrote large-scale RTL in **SystemVerilog** for a Artix-7 FPGA to implement a multi-oscillator audio synthesizer; automated the verification and synthesis flow and validated the design with a comprehensive **DPI-C** testbench.

FPGA UART ALU January 2025

- Developed a custom variable-length packet parsing state machine to drive a 32-bit ALU via UART, enabling real-time verification with Python.

Relevant Coursework

Graduate (M.S. in CSE)

- Formal Methods (CSE 216)
- Advanced VLSI Design (CSE 222A)
- Brain-Inspired ML (CSE 210)

Undergraduate (B.S. in CE)

- Logic Design with Verilog (CSE 125)
- Embedded System Design (CSE 121)
- Probability & Statistics (CSE 107)

Honors, Awards, and Fellowships

[A4]	2026	TLC Graduate Pedagogy Fellowship – Pedagogical Leadership Certificate, Teaching & Learning Center, University of California, Santa Cruz
[A3]	2025	Impact Argonne Award: Energy Secretary Visit, Argonne National Lab
[A2]	2021-2025	Dean's Honors, University of California, Santa Cruz
[A1]	2021	UC Regents Scholar, Regents of the University of California

Technical Reports

[R1] C. Baker, R. Huang, W. Ali, A. Muddamsetti, *Optimal Cell Resizing in OpenROAD-flow-scripts Using Optimization Modulo Theories*, Baskin School of Engineering, University of California, Santa Cruz, Santa Cruz, California, 2025.

[R2] Goberville, Nick and Huang, Raphael, *Modernizing LARS: Open-source, Modular, and Programmatically Flexible Scenario Replay*, Argonne National Lab, Science Undergraduate Laboratory Internships (SULI), Lemont, Illinois, 2025.

[R3] R. Huang, D. Lee, C. Baker, V. Harkins, and G. Austin, *PillPal: A Smart Medication Dispensing System*, Baskin School of Engineering, University of California, Santa Cruz, Santa Cruz, CA, 2025.

[R4] R. Huang, C. Baker, M. Frank, *LiveLLNN: Reconfigurable, Heterogeneous Lookup Table-Based Logical Neural Networks*, Santa Cruz, California, 2026.